

A CORE FOR EACH CHORE

Dual- and quad-core processors will soon be old hat, with industry majors working towards six-core desktop CPUs and eight-core and twelve-core server processors, not to forget the prototypes with dozens of cores. Ongoing research also promises highly scalable and customisable heterogeneous multi-core products in the future, complemented with the right parallel programming tools to take advantage of these capabilities



Intel Core i5 processor

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Have you ever wished you had four hands so you could work faster? A similar sentiment amongst computer engineers was what led to the development of multi-core processors more than a decade ago.

A multi-core processor is essentially a chip that has more than one processor (or core) attached to it. By engaging these multiple cores using the right software, a computer can do

more in less time.

There is a great demand for multi-core processors across various application domains ranging from general-purpose and embedded computing to networking, digital signal processing (DSP) and graphics. Over the past few years, multi-cores have become the norm, with dual- and quad-core processors being taken for granted even in the desktop and mobile computing spaces.

Of course, we all know that in the computing world, users are never satis-

fied. However fast their computers are, they want more speed. When they find four cores mundane, they will have six-core processors (Intel has already demonstrated one chip), but soon even that will not be enough. Every need will be met by the industry, with more research, more innovations and more products. It is an unending cycle, and no technology is permanent.

Let us therefore stop at this moment in time and capture the current state of affairs in multi-core processors. Well, the situation here is quite similar to the Cola Wars, with Intel and AMD trying to outdo each other with more cores.

A six-mare chariot to pull your computer

Six- and even eight-core processors *per se* are not news any more, with server-focused products such as Intel's Xeon and AMD's Opteron being quite popular. However, it is interesting to note that Intel recently demonstrated a six-core processor for the desktop segment that is currently being ruled by dual- and quad-core processors. This processor might be launched sometime in March this year.

The processor, codenamed Gulf-town, will use Intel's 32-nanometre (nm) fabrication technology and Westmere architecture, which means it will also feature some of the interesting power-saving techniques used in Intel's 45nm products. The power management functions of this architecture will ensure power savings by limiting